

Integrated User Association, Computation Offloading, Resource Allocation, and UAV Trajectory Control Against Jamming for UAV-Based Wireless Networks

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Abstract—In this paper, we address optimum design of uncrewed aerial vehicle (UAV)–based wireless networks with a focus on computation offloading in the presence of an active aerial attacker. Our design aims to minimize the maximum computation time among the tasks of ground users while satisfying the energy consumption requirements. To this end, we propose a joint optimization problem of partial computation offloading, ground user association, multiple UAVs trajectory control, computation resource, and sub-channel assignment. To tackle the underlying non-convex mixed-integer nonlinear optimization problem, we use the alternating optimization approach to iteratively solve the five sub-problems, namely, user-UAV association, user scheduling, partial offloading control and bit allocation over time slots, computation resource and sub-channel assignment, and UAV trajectory control until convergence. Moreover, the successive convex approximation method is employed to solve the non-convex sub-problems and improve the resilience of the system against jammer attacks. Additionally, we propose low-complexity algorithms to solve the involved sub-problems. Via extensive numerical studies, we illustrate the effectiveness of our proposed design compared to baselines under the impact of an aerial attacker.

Index Terms—UAV trajectory, mobile edge computing, jamming, non-convex mixed-integer nonlinear optimization.

I. INTRODUCTION

FOR future wireless communication networks, several promising technologies are being considered, including uncrewed aerial vehicle (UAV) communications and mobile edge computing (MEC) [1], [2]. Due to their low cost, high

altitude, and ease of deployment, the integration of UAV and MEC systems offers a flexible solution for scenarios where terrestrial MEC networks may be impractical, such as in disaster-stricken areas, battlefields, and emergency zones. This integration also enables the provision of MEC services with low latency, reduced energy consumption, and enhanced flexibility [3], [4].

On the other hand, the integration of UAV and MEC technologies rises up numerous challenges related to computation offloading and resource allocation. Recently, several research work have addressed these challenges [5], [6], [7], [8], [9], [10], [11], wherein computation tasks have been offloaded from ground users (GUs) to UAV-mounted edge servers and satellites. Specifically, Feng et al. [5] considered a multi-user MEC system and optimized GU-UAV association and task partitioning to achieve minimum average latency for all GUs. Meanwhile, Tang et al. [6] studied GU-UAV association and task partitioning to achieve minimum packet delay. Maximizing the number of tasks that satisfy delay constraints has been addressed in [7]. In addition, the objective of minimizing the long-term time averaged network operation cost was considered in [8], while minimizing the expected total latency and minimizing the difference between the allocated and required data rates were investigated in [9] and [10], respectively. Furthermore, a flexible resource scheduling framework was proposed in [11] with the objective of minimizing energy consumption of the MEC system. However, these work only optimized the UAV placement or assumed that the UAVs' trajectories are pre-determined.

Due to the broadcast nature and openness of aerial-ground networks, malicious devices can intentionally send jamming signals to hinder transmission and degrade communication performance [12]. Some researchers have addressed this issue [13], [14], [15], [16], [17]. In particular, Han et al. [13] considered a coordinated satellite-UAV communication system, where the UAV performs the environmental reconnaissance task with the assistance of satellites in a hostile jamming environment. Wang et al. [14] proposed an aerial bridge scheme to secure massive data transmissions in the presence of eavesdroppers. Shi et al. [15] studied the system with satellites sending jamming signals to the eavesdroppers. This design

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aims to maximize the system secrecy energy efficiency via jointly optimizing transmit power allocation and user association. Moreover, an efficient secure communication scheme was proposed in [16] to maximize secure computing capacity by jointly optimizing the transmit power, time slot allocation factor, computation optimization and UAV trajectory. Meanwhile, maximization of the fairness in achievable secrecy rate among GUs was studied in [17] by considering joint UAV trajectory, transmission power, and scheduling time for task offloading. However, the joint problem of user association and computation resource allocation under the jamming attacks has not yet been considered in these studies. Motivated by the above facts, our work aims to study computation offloading, resource allocation, and UAV trajectory control against jamming in UAV-based wireless networks as a whole.

A. Related Work

A summary of recent related work is given in Table I, where some of the designs have simultaneously considered of different key aspects including user energy constraint, bandwidth allocation, UAV trajectory optimization, parallel offloading control, user association, and jammer attack. Clearly, Table I shows that our current work provides the most comprehensive design.

In the above-listed work, several studies have focused on exploiting the benefits of offloading and resource allocation in multi-UAV and base station (BS) aerial-ground cooperative MEC systems [18], [19], [20], [21], [22]. In particular, UAV placement and transmission power allocation were studied in [18], where UAVs were served as relays between users and remote cloudlet. Ei et al. [19] studied a multi-UAV-assisted-two-stage MEC system, where users can offload their data to UAVs or to the ground BS. Meanwhile, both the centralized and distributed frameworks were proposed in [20] for a UAV and BS collaboration MEC system, where the UAV not only provided MEC services for the users but also for the neighboring cells and can relay computation tasks to the MEC-enabled BS as well. In [21], the energy consumption in a multi-UAV and BS hybrid-enabled MEC system was minimized by jointly optimizing the UAV trajectories, task associations, computation, and transmission resources. Moreover, the joint optimization of user association, CPU-cycle frequency, transmit powers, offloading time, and UAVs' flying trajectories to maximize the total computation bits was studied in [22]. However, these existing work have not considered the bandwidth allocation and they only tackled the binary task assignment or task partitioning between the local device and the MEC or cloud servers.

The joint optimization of sub-channel selection, power control, and UAV deployment was studied in [23], where the design objective is to maximize the system's energy efficiency. Meanwhile, a system with a single satellite, a single UAV, and multiple small cells was studied in [24], whose design aims to maximize the sum rate of the small cells by jointly optimizing the user association, sub-channel and power allocation considering the maximum delay constraint. Mao et al. [25] considered minimization of the maximum delay experienced by different GUs by jointly optimizing user association, task

assignment, power control, continuous bandwidth allocation, computation resource, and UAV placement. Nguyen et al. [26] studied the joint optimization problem of user association, task assignment, continuous bandwidth allocation, computation resource, and UAV trajectory control to achieve minimum weighted energy consumption of GUs and UAVs. However, these designs have not considered the presence of an active jammer in non-terrestrial networks.

To address the challenges of jammer attacks in UAV-based wireless networks, several studies have been conducted [27], [28], [29], [30], [31], [32]. In particular, Ding et al. [27] studied maximization of secure computation efficiency by optimizing user binary offloading decision and resource management. Additionally, the maximization of minimum secure computing capacity was investigated in [28] by jointly optimizing the communication resources, computation resources, and UAVs' trajectories. Besides, the optimization of UAV trajectory and parallel offloading control to minimize transmission delay was addressed in [29]. Furthermore, Liu et al. [30] investigated the minimization of task completion time by jointly optimizing UAV trajectories, user associations, and channel selections in the presence of jamming. The problem of hover point selection and resource management under dynamic jamming attacks, with the objective of maximizing overall communication and computing efficiency, was studied in [31]. Meanwhile, the objective of minimizing the weighted system cost, including latency and energy consumption, was considered in [32] by jointly optimizing UAV trajectory, offloading strategy, and transmit power. However, the joint problem of user association, energy, and computation resource allocation in the presence of jammer attacks has not yet been explored in these studies.

It should be mentioned that several key design aspects were not addressed satisfactorily in the aforementioned existing work. Especially partial offloading for computation load balancing among GUs, edge, and cloud servers under the impact of radio resource allocation and UAV trajectory control against jamming attacks has not been studied in the literature. In [33], we studied single UAV trajectory control with computation offloading against jamming without considering the hybrid BS and cloud computing, user association, and flying jammer. Therefore, by addressing the gap in the literature, this paper proposes a comprehensive design framework for hybrid edge-cloud-based multi-UAV networks and provide novel solutions to optimize the system's performance in the presence of an aerial jammer. To the best of our knowledge, none of the existing work has studied the joint user association and UAV trajectory control with computation offloading and resource allocation against an aerial jammer in UAV-based wireless networks.

B. Contributions

The main contributions can be summarized as follows:

- We study the joint user association, partial computation offloading, UAV trajectory control, computation resource, and sub-channel assignment in UAV-based wireless networks under the impact of an aerial jammer. This design

TABLE I

RELATED WORK ON COMPUTATION OFFLOADING AND RESOURCE ALLOCATION AGAINST JAMMER ATTACK IN UAV-BASED WIRELESS NETWORKS

Ref.	Objective	Enabled BS-Cloud Computing	Energy Constraint	Bandwidth Allocation Optimization	UAV Trajectory Optimization	Parallel Offloading Control	User Association	Jammer Attack
[18]	Minimize energy consumption	✓	✓				✓	
[19]	Minimize energy consumption	✓					✓	
[20]	Maximize the utility of the proposed UAV-enabled MEC network	✓					✓	
[21]	Minimize energy consumption	✓			✓		✓	
[22]	Maximize the computation bits	✓	✓		✓		✓	
[23]	Maximize the system energy efficiency			Sub-channel assignment			✓	
[24]	Maximize the sum rate of the small cell		✓	Sub-channel assignment		✓	✓	
[25]	Minimize the maximum computation delay among IoT devices		✓	Continuous bandwidth allocation		✓	✓	
[26]	Minimize weighted energy consumption of GUs and UAVs			Continuous bandwidth allocation	✓	✓	✓	
[27]	Maximize the secure computation efficiency						✓	✓
[28]	Maximize the minimum secure computing capacity				✓		✓	✓
[29]	Minimize the transmission delay				✓	✓		✓
[30]	Minimize task completion time			Sub-channel assignment	✓		✓	✓
[31]	Maximize the tasks processing energy efficiency				✓			✓
[32]	Minimize weighted system cost of latency and energy consumption				✓	✓	✓	✓
[33]	Minimize the maximum processing time among GUs		✓	Continuous bandwidth allocation	✓	✓		✓
This Work	Minimize the maximum processing time among GUs	✓	✓	Sub-channel assignment	✓	✓	✓	✓

aims to minimize the maximum computation time among the tasks of GUs while satisfying the energy consumption constraints.

- We formulate a non-convex mixed-integer nonlinear optimization problem and explore the alternating optimization approach to solve it. In particular, the successive convex approximation (SCA) method is employed to solve the computation resource and sub-channel assignment and the UAV trajectory control sub-problems.
- To reduce computational complexity, we propose heuristic algorithms to solve the sub-problems of GU-UAV association, user scheduling, partial computation offloading, computation resource, and sub-channel assignment.
- Numerical results are presented to show the impacts of different parameters and the importance of trajectory optimization on the achievable performance, thereby showing the resilience of the proposed approach in the presence of aerial attackers.

C. Organization

The remainder of this paper is organized as follows. Section II presents the system model and problem formulation. In Section III, we develop in detail our integrated approach to solve the complex non-convex mixed-integer nonlinear optimization problem. Section IV presents numerical studies for evaluating the performance of the developed algorithms. Finally, Section V concludes this work.

II. SYSTEM MODEL

We consider a UAV-based wireless network model with anti-jamming as illustrated in Fig. 1, where the ground network layer includes K GUs, the aerial network layer employs M UAVs, and the ground BS is connected to a distant cloud

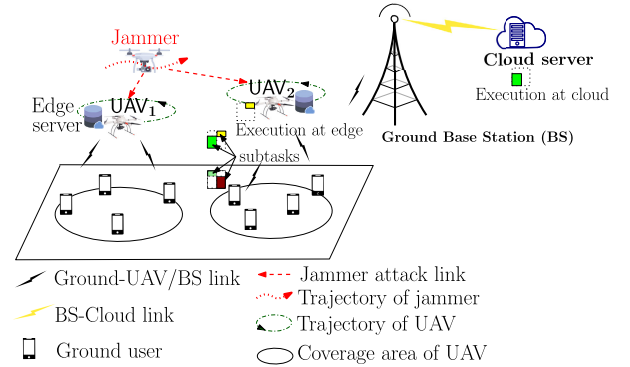


Fig. 1. Proposed multi-UAV wireless network model with anti-jamming.

server. This model reflects practical scenarios such as temporary events where the GUs have tasks that require immediate computation. In addition, a malicious aerial attacker device (denoted as a jammer) can jam the full bandwidth of the network to disturb the legitimate communications between GUs and UAVs. Particularly, the fixed wing UAV jammer is considered in our system.

We denote the sets of UAVs and GUs as $\mathcal{M} = \{1, \dots, M\}$, and $\mathcal{K} = \{1, \dots, K\}$, respectively. All GUs are assumed to have fixed horizontal coordinates of $\mathbf{g}_k^0 = (x_k^0, y_k^0), \forall k \in \mathcal{K}$. Besides, we assume that UAVs fly at a fixed altitude H over a flight period of $T > 0$ seconds. The flight period is divided into N time slots, whose set is denoted as $\mathcal{N} = \{1, \dots, N\}$. In addition, the attacker (jammer) flies at a fixed altitude H^j over a flight period of T with the horizontal coordinates denoted as $\mathbf{g}^j[n] = (x^j[n], y^j[n]), \forall n \in \mathcal{N}$. Furthermore, the horizontal coordinate of the BS can be denoted by $\mathbf{r}^b = (x^b, y^b)$ and its fixed altitude is H^b .

We assume that the communication links from the GUs to UAVs are dominated by the line-of-sight (LoS) propagation, while the ground-to-ground transmission links between the BS and GUs are modeled as Rayleigh fading links. Therefore, we assume that the GU-UAV channels change over time slots due to different positions of UAVs, while the GU-BS channels are fixed during flight.

The details of channel state information (CSI), fading channels, and dynamic systems have been studied in [34], [35], [36], and [37]. Moreover, UAV CSI estimation techniques can be used to obtain and predict the state of wireless communication channels in UAV networks. However, these techniques face unique challenges in UAV communications, such as high mobility, three-dimensional (3D) spatial dynamics, and varying environmental conditions. To address these challenges, several research directions have been proposed, such as deploying edge computing platforms to support real-time, artificial intelligence (AI)-driven CSI estimation for UAV networks, leveraging joint sensing and communication frameworks to improve CSI accuracy by integrating environmental awareness, or using cooperative estimation techniques where multiple UAVs share CSI information to enhance the network's overall performance. Therefore, the details of UAV CSI estimation, fading channels, and dynamic systems for UAV-based wireless networks will be explored and thoroughly investigated in future work.

A. Computation Task Model

Partial computation offloading is considered where each GU may partition its computation task into three sub-tasks to be processed locally, at the UAV-mounted edge server, and at the cloud server, respectively. The data related to the third sub-task must be transmitted from the GU to the cloud server through the BS as in [5] and [25]. Since the computation results produce much smaller data sets, we neglect the download time of the computation results for the second and third sub-tasks.

Suppose that GU k has a computation task denoted by $U_k = (f_k, s_k, r_k)$, where f_k denotes the computation demand expressed by the number of central process unit (CPU) cycles per second (CPU cycles/second), s_k (in bits) represents the size of input raw data, and r_k (CPU cycles/bit) denotes the computation resource required for 1-bit input data.

The task processing time for GU k can be written as:

$$T_k = \max \left\{ T_k^{\text{lo}}, T_k^{\text{uv}}, T_k^{\text{os}} \right\}, \quad (1)$$

where T_k^{lo} , T_k^{uv} , and T_k^{os} represent the task execution time (in addition to possibly offloading time) at the GU, UAV-mounted edge server, and cloud server, respectively.

Let α_k^{lo} and α_k^{uv} , ($0 \leq \alpha_k^{\text{lo}}, \alpha_k^{\text{uv}} \leq 1$) denote the fractions of input data to be processed locally at GU k and to be offloaded and processed at the UAV-mounted edge server, respectively. Hence, $(1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{uv}})$ represents the fraction of input data to be offloaded and processed at the distant cloud server.

B. UAV Trajectory Control

The UAV's energy is assumed to be sufficiently large. This assumption is supported by [38] where a UAV equipped

with a 3-cell, 3250mAh, and 11.1V LiPo battery can have a flight time of about 20 minutes. However, the UAV flight period T should be chosen to achieve a good balance between the user throughput and access delay [39]. The horizontal coordinates of UAV m in time slot n is denoted as $\mathbf{u}_m[n] = (x_m^u[n], y_m^u[n])$. We assume that each UAV must come back to its initial position at the end of the flight period. This constraint can be formulated as follows:

$$\mathbf{u}_m[1] = \mathbf{u}_m[N], \quad \forall m \in \mathcal{M}. \quad (2)$$

In addition, the slot interval $\delta = \frac{T}{N}$ is chosen to be sufficiently small so that the UAVs' locations are within a bounded neighborhood in each time slot even at the maximum flight speed $V_{\max}$. Hence, the trajectory of the UAVs must meet the following constraints:

$$\|\mathbf{u}_m[n+1] - \mathbf{u}_m[n]\|^2 \leq D_{\max}^2, \quad n = 1, \dots, N-1, \quad (3)$$

$$\|\mathbf{u}_m[n] - \mathbf{u}_z[n]\|^2 \geq d_{\min}^2, \quad \forall n, m, z \neq m, \quad (4)$$

$$\|\mathbf{u}_m[n] - \mathbf{g}^j[n]\|^2 \geq d_{\min}^2, \quad \forall n, m, \quad (5)$$

where $\|\cdot\|$ denotes the Euclidean norm, $D_{\max} \triangleq V_{\max}\delta$ is the maximum horizontal distance that the UAV can travel in each time slot, $d_{\min}$ denotes the minimum allowed value for UAV-UAV and UAV-jammer distances to ensure collision avoidance.

C. GU-UAV Association

Let $\omega_{k,m}[n]$ denote binary association variables, where $\omega_{k,m}[n] = 1$, if GU k is served by UAV m in time slot n and $\omega_{k,m}[n] = 0$, otherwise. The requirement for the association is that each GU can just offload its computation sub-task to at most one UAV in each time slot. This constraint can be expressed as:

$$\sum_{m=1}^M \omega_{k,m}[n] \leq 1, \quad \forall k, n. \quad (6)$$

D. Sub-Channel Assignment

We assume that orthogonal frequency division multiple access (OFDMA) is used between the GUs and UAVs (or BS). Let $\mathcal{C} = \{1, \dots, C\}$, $\mathcal{C}^A = \{1, \dots, C^A\}$, and $\mathcal{C}^B = \{1, \dots, C^B\}$ be the set of all sub-channels, the set of sub-channels for access links between the GUs and the UAVs, and the set of sub-channels between the GUs and the BS, respectively. Obviously, $\mathcal{C} = \mathcal{C}^A \cup \mathcal{C}^B$. Let W (MHz) denote the bandwidth of each sub-channel.

As assumed above, the channels between the GUs and BS change over time slots, then the sub-channels should be assigned over each time slot. Hence, we define sub-channel assignment variables for GU-UAV links as follows:

$$\phi_{k,c}^u[n] = \begin{cases} 1 & \text{if sub-channel } c \text{ is assigned for GU } k \\ & \text{served by UAV in time slot } n, \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

It is assumed that each sub-channel c can be assigned to support at most one GU-UAV in each time slot. This constraint can be expressed as:

$$\sum_{k \in \mathcal{K}} \omega_{k,m}[n] \phi_{k,c}^u[n] \leq 1, \quad \forall m, n, c \in \mathcal{C}^A. \quad (8)$$

This constraint justifies the need to distinguish between sub-channels, even though they all have the same bandwidth.

Furthermore, the channels between the GUs and BS are fixed during flight time. Therefore, we define variables to assign the sub-channels for GU-BS links as follows:

$$\phi_{k,c}^b = \begin{cases} 1 & \text{if sub-channel } c \text{ is assigned for GU } k \\ & \text{served by the BS,} \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

In addition, each sub-channel c can be assigned to support at most one GU-BS link. This constraint can be expressed as:

$$\sum_{k \in \mathcal{K}} \phi_{k,c}^b \leq 1, \quad \forall c \in \mathcal{C}^B. \quad (10)$$

E. Communication Model

1) *GU-BS Communication Model*: The distance from GU k to the BS can be calculated as $d_{k,0} = \sqrt{(H^b)^2 + \|\mathbf{r}^b - \mathbf{g}_k^0\|^2}$. The channel power gain from GU k to the BS can be calculated as

$$g_k^B = \rho_0 |h_0|^2 d_{k,0}^{-2} = \frac{\rho_0 |h_0|^2}{(H^b)^2 + \|\mathbf{r}^b - \mathbf{g}_k^0\|^2}, \quad (11)$$

where ρ_0 is the channel power gain at the reference distance of 1 m, h_0 denotes a Gaussian random variable with zero mean and unit variance. Hence, the corresponding received signal-to-noise ratio (SNR) at the BS can be expressed as

$$\gamma_k^B = \frac{P_k^B g_k^B}{\sigma^2}, \quad (12)$$

where P_k^B denotes the transmit power of GU k to the BS on each sub-channel, and σ^2 represents the power of the additive white Gaussian noise (AWGN) at the receiver. Therefore, the corresponding rate denoted by $R_{k,c}^B$ from GU k on sub-channel c can be expressed as

$$R_{k,c}^B = \phi_{k,c}^b W \log_2 (1 + \gamma_k^B). \quad (13)$$

2) *GU-UAV Communication Model*: The channel power gain from GU k to UAV m in time slot n is assumed to follow the free-space path loss model. It can be expressed as:

$$g_{k,m}[n] = \frac{\rho_0}{H^2 + \|\mathbf{u}_m[n] - \mathbf{g}_k^0\|^2}, \quad (14)$$

Hence, the achievable rate of the uplink transmission from GU k to the associated UAV m on sub-channel c in time slot n denoted by $R_{k,m,c}^u[n]$ can be modeled as:

$$R_{k,m,c}^u[n] = \omega_{k,m}[n] \phi_{k,c}^u[n] W \log_2 \left(1 + \frac{P_k^u g_{k,m}[n]}{I_{k',k,m,c}^u[n] + I_m^j[n] + \sigma^2} \right), \quad (15)$$

where σ^2 denotes the power density of the additive white Gaussian noise (AWGN) at the receiver, $I_m^j[n] = P^j g_m^j[n]$ represents the interference at UAV m due to the jammer attacks

in time slot n , and $I_{k',k,m,c}^u[n] = \sum_{k' \neq k} \phi_{k',c}^u[n] P_{k',c}^u g_{k',m}[n]$ represents the interference at UAV m on the sub-channel c due to the transmissions of other GUs in time slot n on this channel. In addition, P_k^u represents the uplink transmit power of GU k on each sub-channel. Moreover, P^j and $g_m^j[n]$ represent the transmit power of the jammer and channel power gain from the jammer to UAV, which can be calculated as:

$$g_m^j[n] = \frac{\rho_0}{(H - H^j)^2 + \|\mathbf{u}_m[n] - \mathbf{g}^j[n]\|^2}. \quad (16)$$

F. Task Execution Time and Energy Consumption

1) *Local Computing*: The local task execution time at GU k can be expressed as:

$$T_k^{\text{lo}} = \frac{\alpha_k^{\text{lo}} s_k r_k}{f_k}. \quad (17)$$

The energy consumption due to local task execution is:

$$E_k^{\text{lo}} = \kappa \alpha_k^{\text{lo}} s_k r_k (f_k)^2, \quad (18)$$

where κ is the effective switched capacitance depending on the chip architecture [40].

2) *UAV-Mounted Edge Server Computing*: Let $\tau_{k,c}^u[n]$ denote the number of offloading bits from GU k to the associated UAV on sub-channel c over time slot n . Besides, let us denote the computing capacity of UAV m to handle the sub-task offloaded from GU k at time slot n by $f_k^u[n]$ (CPU cycles/second). Then, the task execution time at the associated UAV in time slot n can be computed as:

$$T_k^{\text{eu}}[n] = \frac{r_k}{f_k^u[n]} \sum_{c \in \mathcal{C}^A} \{ \phi_{k,c}^u[n] \tau_{k,c}^u[n] \}. \quad (19)$$

The required number of time slots to completely offload the sub-task from GU k to the UAVs can be calculated as:

$$N_k^u = \sum_{n \in \mathcal{N}} \left\lceil \max_{c \in \mathcal{C}^A} \left\{ \sum_{m \in \mathcal{M}} \frac{\omega_{k,m}[n] \phi_{k,c}^u[n] \tau_{k,c}^u[n]}{R_{k,m,c}^u[n]} \right\} \frac{1}{\delta} \right\rceil, \quad (20)$$

where $\lceil \cdot \rceil$ denotes the ceiling function.

We now introduce binary user scheduling variables $\theta_k[n]$, where $\theta_k[n] = 1$ if GU k is scheduled to transmit to its associated UAV in time slot n and $\theta_k[n] = 0$, otherwise. We need to impose the following constraints on the user scheduling decisions:

$$\sum_{m \in \mathcal{M}} \sum_{t=0}^{N_k^u-1} \theta_k[n+t] \omega_{k,m}[n+t] = N_k^u, \quad \forall k, \quad n \in \{1, \dots, N - N_k^u\}. \quad (21)$$

The offloading time from GU k to UAV m in time slot n can be expressed as:

$$T_k^{\text{ou}}[n] = \max_{c \in \mathcal{C}^A} \left\{ \sum_{m \in \mathcal{M}} \frac{\omega_{k,m}[n] \theta_k[n] \phi_{k,c}^u[n] \tau_{k,c}^u[n]}{R_{k,m,c}^u[n]} \right\}. \quad (22)$$

The energy consumption for this data transmission can be calculated as:

$$E_k^{\text{ou}}[n] = \sum_{c \in \mathcal{C}^A} \sum_{m \in \mathcal{M}} \frac{\omega_{k,m}[n] \theta_k[n] \tau_{k,c}^u[n] P_k^u}{R_{k,m,c}^u[n]}. \quad (23)$$

Moreover, we assume that the partial sub-task from GU k is offloaded and processed completely on each sub-channel at the associated UAV in each time slot. Then we have the following constraint:

$$T_k^{uv}[n] = T_k^{eu}[n] + T_k^{ou}[n] \leq \delta, \quad \forall k, m, n. \quad (24)$$

The total processing time at the UAVs to serve GU k can be written as:

$$T_k^{uv} = \delta N_k^u. \quad (25)$$

3) *BS-Cloud Server Computing*: Similar to the UAV-mounted edge server computing, let $\tau_{k,c}^b$ denote the number of offloading bits from GU k to the BS on sub-channel c . Due to higher computation resource for execution of the offloaded sub-tasks at the cloud server, we omit the processing time at the cloud server and we also ignore the cloud energy consumption involved in task execution and transmission of the computation results from the cloud server to GUs. Hence, the offloading time from GU k to the BS is given by:

$$T_k^{ob} = \max_{c \in C^B} \left\{ \frac{\phi_{k,c}^b \tau_{k,c}^b}{R_{k,c}^B} \right\}, \quad \forall k. \quad (26)$$

Then, the offloading time from GU k to the cloud server through the BS can be calculated as:

$$T_k^{os} = T_k^{ob} + T_k^{td}, \quad (27)$$

where T_k^{td} is the transmission delay from the BS to the cloud server [41] and can be calculated as:

$$T_k^{td} = \frac{(1 - \alpha_k^{lo} - \alpha_k^{uv})s_k}{\vartheta}, \quad (28)$$

where ϑ denotes the communication capacity of the link from the BS to the cloud server.

Furthermore, the energy consumption for transmitting the offloaded data from the BS to the cloud server can be calculated as follows:

$$E_k^{os} = \sum_{c \in C^B} \frac{\phi_{k,c}^b \tau_{k,c}^b P_k^B}{R_{k,c}^B}. \quad (29)$$

G. Problem Formulation

The total energy consumption of GU k to completely process its computation task can be expressed as:

$$E_k = E_k^{lo} + \sum_{n \in \mathcal{N}} E_k^{ou}[n] + E_k^{os}, \quad \forall k. \quad (30)$$

For convenience, we gather different decision variables as GU-UAV association $\Omega = \{\omega_{k,m}[n], \forall k, m, n\}$, user scheduling $\Theta = \{\theta_k[n], \forall k, n\}$, partial offloading control $\mathbf{A} = \{\alpha_k^{lo}, \alpha_k^{uv}, \forall k\}$, bit allocation $\Upsilon = \{\tau_{k,c}^u[n], \tau_{k,c}^b, \forall k, n, c\}$, sub-channel assignment $\Phi = \{\phi_{k,c}^u[n], \phi_{k,c}^b, \forall k, c, n\}$, computation resource allocation $\mathbf{F} = \{f_k^u[n], \forall k, n\}$, and UAV trajectory control $\mathbf{U} = \{\mathbf{u}_m[n], \forall m, n\}$.

The task processing time T_k in (1) is a nonlinear function with respect to (w.r.t) the decision variables Ω , Θ , $\mathbf{A}$, Υ , Φ , $\mathbf{F}$, and $\mathbf{U}$. We introduce the function

$\eta(\Omega, \Theta, \mathbf{A}, \Upsilon, \Phi, \mathbf{F}, \mathbf{U}) = \max_{k \in \mathcal{K}} T_k$ as the maximum processing time among all GUs. Then, our optimization problem becomes equivalent to minimizing $\eta(\Omega, \Theta, \mathbf{A}, \Upsilon, \Phi, \mathbf{F}, \mathbf{U})$, which is more tractable. Then, the optimization problem can be formulated as

(P):

$$\min_{\Omega, \Theta, \mathbf{A}, \Upsilon, \Phi, \mathbf{F}, \mathbf{U}, \eta} \eta \quad (31)$$

s.t. constraints (2) – (6), (8), (10), (21), (24), (26),

$$T_k^{lo} \leq \eta, \quad \forall k, \quad (31a)$$

$$T_k^{os} \leq \eta, \quad \forall k, \quad (31b)$$

$$T_k^{uv} \leq \eta, \quad \forall k, \quad (31c)$$

$$E_k \leq E_k^{\max}, \quad \forall k, \quad (31d)$$

$$\sum_{m \in \mathcal{M}} \sum_{n \in \mathcal{N}} \sum_{c \in \mathcal{C}} \omega_{k,m}[n] \theta_k[n] \tau_{k,c}^u[n] = \alpha_k^{uv} s_k, \quad \forall k, \quad (31e)$$

$$\sum_{c \in C^B} \tau_{k,c}^b = (1 - \alpha_k^{lo} - \alpha_k^{uv})s_k, \quad \forall k, \quad (31f)$$

$$\sum_{k \in \mathcal{K}} \omega_{k,m}[n] \theta_k[n] f_k^u[n] \leq F_m^{\max}, \quad \forall m, n, \quad (31g)$$

$$\omega_{k,m}[n], \theta_k[n] \in \{0, 1\}, \quad \forall k, m, n, \quad (31h)$$

$$0 \leq \alpha_k^{lo}, \alpha_k^{uv} \leq 1, \quad \forall k, \quad (31i)$$

$$\phi_{k,c}^u[n], \phi_{k,c}^b \in \{0, 1\}, \quad \forall k, c, n, \quad (31j)$$

$$f_k^u[n], \tau_{k,c}^u[n], \tau_{k,c}^b \geq 0, \quad \forall k, m, n, \quad (31k)$$

where constraints (31a)-(31c) and (24) capture the delay requirements for the GUs. Constraints (31c)-(31h) and (6) describe the binary GU-UAV association variables. Constraints (8) capture the sub-channel assignment for transmission between the GUs and UAVs while constraints (31g) and (31k) present the UAVs' computation constraints where $F_m^{\max}$ denotes the maximum computation resource of UAV m . The partial offloading and bit allocation variables are described in the constraints (31e), (31i), and (31k). Constraints (2)-(5) describe the constraints of UAV trajectory control. It can be seen that the objective and constraint functions (31a)-(31d) are nonlinear and integer decision variables are involved in (31h) and (31j) for the GU-UAV association and sub-channel assignment, respectively. Hence, problem (31) is a mixed-integer nonlinear optimization problem (MINLP) problem, which is difficult to solve optimally. In the following section, we describe how to compute good feasible solutions to this problem.

III. PROPOSED ALGORITHMS

We adopt the alternating optimization approach to solve problem (31) where we iteratively optimize each set of variables given the values of other variables in the corresponding sub-problems until convergence. We describe how to solve these different sub-problems in the following.

A. GU-UAV Association Sub-Problem

Given $\{\Upsilon, \mathbf{A}, \mathbf{F}, \Phi, \mathbf{U}\}$, the GU-UAV association sub-problem to optimize Ω can be formulated as:

$$\begin{aligned} (\mathbf{P1}): \min_{\Omega, \eta} \quad & \eta \\ \text{s.t.} \quad & (31c)-(31h), (6), (24). \end{aligned} \quad (32)$$

1) *Optimization Approach*: It can be verified that the objective function and all constraints of problem (32) are linear w.r.t Ω . Hence, it is a standard mixed integer linear program (MILP), which can be solved efficiently by using the CVX-Gurobi solver [42]. Since CVX solver invokes the interior-point method to solve the underlying problem (32), the involved complexity is $\mathcal{O}(t_1^{1/2}(t_1 + t_2)t_2^2)$, where t_1 is the number of inequality constraints, t_2 denotes the number of variables, and $\mathcal{O}$ denotes the big-O notation [43]. Hence, the complexity of this approach becomes $\mathcal{O}(MNK^{\frac{7}{2}})$.

2) *Iterative GU-UAV Association (IGA)*: We propose a heuristic low-complexity algorithm for the GU-UAV association sub-problem as described in Algorithm 1.

We first assume in step 2 that each GU k is initially associated with the UAV providing the highest average received signal strength (RSS) as follows:

$$\omega_{k,m}[n] = \begin{cases} 1 & m = \arg \max_k RSS_{k,m}[n], \\ 0 & \text{otherwise,} \end{cases} \quad (33)$$

where the average RSS for the GU k and UAV m can be expressed as

$$RSS_{k,m}[n](\text{dBm}) = P_k^u(\text{dBm}) + g_{k,m}[n](\text{dBm}), \quad (34)$$

where P_k^u is the transmit power of GU k used in its communication with associated UAV in time slot n on each assigned sub-channel. Then the algorithm enters an iterative association loop where, in each iteration, it starts with each GU k and attempts to re-assign variable $\omega_{k,m}[n]$ to be served by another UAV in each time slot to achieve better processing time for the task of GU k and also ensure the maximum processing time of the system not increasing. These steps are described from step 7 to step 18. After attempting assignment for all GUs, the maximum processing time of the system is updated in step 20, and the algorithm terminates when this value cannot be reduced further. The complexity of Algorithm 1 is $\mathcal{O}(L_1 KMN)$, where L_1 is the number of iterations required for convergence.

B. User Scheduling Sub-Problem

Given $\{\Omega, \Upsilon, \mathbf{A}, \mathbf{F}, \Phi, \mathbf{U}\}$, the user scheduling sub-problem to optimize Θ can be formulated as:

$$\begin{aligned} (\mathbf{P2}): \min_{\Theta, \eta} \quad & \eta \\ \text{s.t.} \quad & (31c)-(31h), (6), (21), (24). \end{aligned} \quad (35)$$

1) *Optimization Approach*: It can be verified that the objective function and all constraints are linear w.r.t binary variables Θ . Hence, the problem is a standard mixed integer linear program (MILP), which can be solved efficiently by using the CVX-Gurobi solver [42]. The computation complexity of this optimization approach is also $\mathcal{O}(MNK^{\frac{7}{2}})$.

Algorithm 1 Iterative GU-UAV Association (IGA)

Require: M UAVs, K GUs, C sub-channels, T flight time;
 1: **Given:** $\Theta, \Upsilon, \mathbf{A}, \mathbf{F}, \Phi, \mathbf{U}$;
Ensure: Min processing time η ;
 2: **Initialization:** Associate $\omega_{k,m}[n]$ based on the RSS method;
 3: **repeat**
 4: **for** $k = 1 : K$ **do**
 5: $\max_{T_k} = T_k$;
 6: $\text{maxtime} = \max_{k \in \mathcal{K}} \{T_k\}$;
 7: **for** $m = 1 : M$ **do**
 8: **for** $n = 1 : N$ **do**
 9: **while** $\omega_{k,m}[n] = 0$ **do**
 10: $\omega_{k,m}[n] = 1$;
 11: **if** $(T_k \leq \max_{T_k})$ and $(\max_{k \in \mathcal{K}} \{T_k\} \leq \text{maxtime})$ and $(E_k \leq E_k^{\max})$ **then**
 12: $\omega_{k,m}[n] = 1$; $\text{maxtime} = \max_{k \in \mathcal{K}} \{T_k\}$;
 13: **else**
 14: $\omega_{k,m}[n] = 0$;
 15: **end if**
 16: **end while**
 17: **end for**
 18: **end for**
 19: **end for**
 20: $\eta = \max_{k \in \mathcal{K}} \{T_k\}$;
 21: **until** Convergence
 22: **Return** η, Ω .

Algorithm 2 User Scheduling Algorithm (GSA)

Require: M UAVs, K GUs, C sub-channels, T flight time;
 1: **Given:** $\Omega, \Upsilon, \mathbf{A}, \mathbf{F}, \Phi, \mathbf{U}$;
Ensure: Min processing time η ;
 2: **repeat**
 3: **for** $k = 1 : K$ **do**
 4: $\max_{T_k} = T_k$;
 5: $\text{maxtime} = \max_{k \in \mathcal{K}} \{T_k\}$;
 6: **for** $m = 1 : M$ **do**
 7: **for** $n = 1 : N$ **do**
 8: **while** $\omega_{k,m}[n] = 1$ **do**
 9: $\theta_k[n] = 1$;
 10: **if** $(T_k \leq \max_{T_k})$ and $(\max_{k \in \mathcal{K}} \{T_k\} \leq \text{maxtime})$ and $(E_k \leq E_k^{\max})$ **then**
 11: $\theta_k[n] = 1$; $\text{maxtime} = \max_{k \in \mathcal{K}} \{T_k\}$;
 12: **else**
 13: $\theta_k[n] = 0$;
 14: **end if**
 15: **end while**
 16: **end for**
 17: **end for**
 18: **end for**
 19: $\eta = \max_{k \in \mathcal{K}} \{T_k\}$;
 20: **until** Convergence
 21: **Return** η, Θ .

2) *User Scheduling Algorithm (GSA)*: A heuristic algorithm is proposed in Algorithm 2 to solve the user scheduling

sub-problem. This algorithm enters an iterative scheduling for each GU k with association $\omega_{k,m}[n] = 1$. The scheduling variable $\theta_k[n] = 1$ to achieve better processing time for T_k and also ensure the maximum processing time of the system not increasing. These steps are mentioned in steps 6 – 17. After that, the maximum processing time of the system is updated in step 19 and the Alg. 2 terminates when this value cannot be reduced further. The complexity of Alg. 2 is $\mathcal{O}(L_2 K M N)$, where L_2 is the number of iterations required for convergence.

C. Partial Offloading Control and Bit Allocation Over Time Slots Sub-Problem

Given $\{\Omega, \Theta, \mathbf{F}, \Phi, \mathbf{U}\}$, the sub-problem optimizing the partial offloading control and bit allocation $\{\mathbf{A}, \Upsilon\}$ can be formulated as:

$$\begin{aligned} \text{(P3): } \min_{\mathbf{A}, \Upsilon, \eta} \quad & \eta \\ \text{s.t.} \quad & (31a)-(31d), (31e), (31i), (31k). \end{aligned} \quad (36)$$

1) *Optimization Approach*: It can be verified that the objective function and all constraints in problem (36) are linear w.r.t $\mathbf{A}$ and Υ . Therefore, it is a linear problem (LP) and can be solved by using the CVX-Gurobi solver [42]. The computation complexity of this optimization approach is $\mathcal{O}(NCK^{\frac{7}{2}})$.

2) *Computation Offloading Algorithm (COA)*: In this section, we propose a heuristic algorithm with lower complexity to solve problem (36). It can be seen that for given variables of GU-UAV association, computation resource, sub-channel assignment, and UAV trajectory control, problem (36) is independent among GUs. If we minimize the processing time for each GU, then the maximum processing time among GUs is minimized. Hence, the objective of this proposed algorithm is to minimize the processing time of each GU k . In view of the constraints in (31i), $0 \leq \alpha_k^{\text{lo}}, \alpha_k^{\text{uv}} \leq 1, \forall k$, the bisection search method is applied to solve this sub-problem. Details of the proposed algorithm are described in Algorithm 3.

Due to the limitation of computation resources at GU, it would take the longest time if the task is locally processed. In particular, the processing time, in this case, is calculated as in step 3, $\eta_k^{\text{lo,up}} = T_k^{\text{lo}}(\alpha_k^{\text{lo,max}})$. Hence, we need to partition the computing at local, edge, or cloud servers. The bisection search loop for each GU is described from steps 6 to 29. For each value of α_k^{lo} , we find the optimal value of α_k^{uv} to achieve the smallest value of processing time for GU k . The bisection search method is also applied to find the solution of α_k^{uv} as described in steps 10 – 21. Specifically, for given each value of α_k^{uv} , the number of offloading bits on each channel in each time slot is calculated as $\tau_{k,c}^{\text{u}}[n] = \alpha_k^{\text{uv}} s_k / (\sum_{c,n} \phi_{k,c}^{\text{u}}[n])$ and $\tau_{k,c}^{\text{b}} = (1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{uv}}) s_k / (\sum_c \phi_{k,c}^{\text{b}})$. The maximum processing time at edge or cloud servers is verified to achieve the smallest value as described in steps 14 – 20.

For each pair of solution of $\{\alpha_k^{\text{lo}}, \alpha_k^{\text{uv}}\}$ in each iteration, we calculate the task processing time η_k^{step} for GU k in step 22. Considering this value is not increasing after each iteration from steps 23 to 28, the bisection search loop from steps 6 to

Algorithm 3 Computation Offloading Algorithm (COA)

Require: M UAVs, K GUs, C sub-channels, T flight time,

```

1: Given:  $\Omega, \Phi, \mathbf{F}, \mathbf{U}$ ;
Ensure: Min processing time  $\eta$ ;
2: for  $k = 1 : K$  do
3:    $\alpha_k^{\text{lo,max}} = 1 \rightarrow \eta_k^{\text{lo,up}} = T_k^{\text{lo}}(\alpha_k^{\text{lo,max}})$ ;
4:    $\alpha_k^{\text{lo,min}} = 0$ ;
5:   Update  $\eta_k^{\text{lb}} = \eta_k^{\text{lo,up}}$ ;
6:   while  $(\alpha_k^{\text{lo,max}} - \alpha_k^{\text{lo,min}}) > \epsilon$  do
7:     Assign  $\alpha_k^{\text{lo}} = (\alpha_k^{\text{lo,max}} + \alpha_k^{\text{lo,min}})/2$ ;
8:     Update  $\alpha_k^{\text{ed,min}} = 0$ ;  $\alpha_k^{\text{ed,max}} = 1 - \alpha_k^{\text{lo}}$ ;
9:     Calculate  $\eta_k^{\text{ed,up}} = T_k^{\text{uv}}(\alpha_k^{\text{ed,max}})$ ;
10:    while  $(\alpha_k^{\text{ed,max}} - \alpha_k^{\text{ed,min}}) > \epsilon$  do
11:      Assign  $\alpha_k^{\text{uv}} = (\alpha_k^{\text{ed,max}} + \alpha_k^{\text{ed,min}})/2$ ;
12:      Calculate  $\tau_{k,c}^{\text{u}}[n] = \alpha_k^{\text{uv}} s_k / (\sum_{c,n} \phi_{k,c}^{\text{u}}[n])$ ;  $\tau_{k,c}^{\text{b}} =$ 
         $(1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{uv}}) s_k / (\sum_c \phi_{k,c}^{\text{b}})$ ;
13:      Calculate  $T_k^{\text{uv}}(\alpha_k^{\text{uv}})$  and  $T_k^{\text{os}}(1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{uv}})$ ;
14:      Update  $\eta_k^{\text{ed,step}} = \max \{T_k^{\text{uv}}(\alpha_k^{\text{uv}}), T_k^{\text{os}}(1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{uv}})\}$ ;
15:      if  $\eta_k^{\text{ed,step}} \leq \eta_k^{\text{ed,up}}$  then
16:         $\alpha_k^{\text{ed,max}} = \alpha_k^{\text{uv}}$ ;
17:         $\eta_k^{\text{ed,up}} = \eta_k^{\text{ed,step}}$ ;
18:      else
19:         $\alpha_k^{\text{ed,min}} = \alpha_k^{\text{uv}}$ ;
20:      end if
21:    end while
22:    Update  $\eta_k^{\text{step}} = \max \{T_k^{\text{lo}}(\alpha_k^{\text{lo}}), T_k^{\text{uv}}(\alpha_k^{\text{uv}}), T_k^{\text{os}}(1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{uv}})\}$ ;
23:    if  $(\eta_k^{\text{step}} \leq \eta_k^{\text{lb}})$  and  $(E_k \leq E_k^{\text{max}})$  then
24:       $\alpha_k^{\text{lo,max}} = \alpha_k^{\text{lo}}$ ;
25:       $\eta_k^{\text{lb}} = \eta_k^{\text{step}}$ ;
26:    else
27:       $\alpha_k^{\text{lo,min}} = \alpha_k^{\text{lo}}$ ;
28:    end if
29:  end while
30: end for
31:  $\eta = \max_{k \in \mathcal{K}} \{T_k\}$ ;
32: Return  $\eta, \mathbf{A}, \Upsilon$ .
```

29 converges to an optimal solution of variables $\{\alpha_k^{\text{lo}}, \alpha_k^{\text{uv}}\}$ with the minimum processing time for GU k .

After finding the solution for all GUs, the maximum processing time among the tasks of GUs is updated in step 31. The bisection search method requires $\log_2 \left(\frac{a^{\text{max}} - a^{\text{min}}}{\epsilon} \right)$ iterations for any variables with the maximum and minimum values of a^{max} and a^{min} under the small difference of step size ϵ . Therefore, the computation complexity of this algorithm is $\mathcal{O} \left(K \log_2 \left(\frac{\alpha_k^{\text{lo,max}} - \alpha_k^{\text{lo,min}}}{\epsilon} \right) \log_2 \left(\frac{\alpha_k^{\text{ed,max}} - \alpha_k^{\text{ed,min}}}{\epsilon} \right) \right)$.

D. Computation Resource and Sub-Channel Assignment Sub-Problem

Given $\{\Omega, \Theta, \mathbf{A}, \Upsilon, \mathbf{U}\}$, the sub-problem optimizing the computation resource and sub-channel assignment $\{\mathbf{F}, \Phi\}$ can

be stated as:

$$\begin{aligned} \text{(P4): } \min_{\mathbf{F}, \Phi, \eta} \quad & \eta \\ \text{s.t.} \quad & \text{(31c)-(31d), (31g), (31j), (31k), (8), (10).} \end{aligned} \quad (37)$$

1) *Optimization Approach*: This is a mixed-integer nonlinear problem. We relax the integer variables of sub-channel assignment Φ into continuous variables. Hence, problem (37) can be formulated as follows:

$$\begin{aligned} \text{(P4.1): } \min_{\mathbf{F}, \Phi, \eta} \quad & \eta \\ \text{s.t. } 0 \leq \phi_{k,c}^u[n], \phi_{k,c}^b \leq 1, \quad & \forall k, c, n, \\ & \text{(31c)-(31d), (31g), (31k), (8), (10).} \end{aligned} \quad (38)$$

This problem is non-convex. We use the SCA method to solve it in the following steps. We first re-write the achievable rate $R_{k,m,c}^u[n]$ in constraints (31c)-(31d) as:

$$R_{k,m,c}^u[n] \geq \omega_{k,m}[n] \theta_k[n] \phi_{k,c}^u[n] W \xi_{k,m,c}[n], \quad (39)$$

where

$$\begin{aligned} & \xi_{k,m,c}[n] \\ & \leq \log_2 \left(1 + \frac{P_k^u g_{k,m}[n]}{\sum_{k' \neq k} \phi_{k',c}^u[n] P_{k'}^u g_{k',m}[n] + I_m^j[n] + \sigma^2} \right). \end{aligned} \quad (40)$$

By applying the first-order Taylor expansion at the given points $\phi_{k,c}^{u,r}[n]$ and $\xi_{k,m,c}^r[n]$ in the r -th iteration of the approximation process, the left-hand-side (LHS) of constraint (39) can be approximated as:

$$\begin{aligned} & \omega_{k,m}[n] \theta_k[n] W \phi_{k,c}^u[n] \xi_{k,m,c}[n] \\ & \geq \frac{1}{4} \omega_{k,m}[n] \theta_k[n] W \left[- \left(\phi_{k,c}^{u,r}[n] + \xi_{k,m,c}^r[n] \right)^2 \right. \\ & \quad + 2 \left(\phi_{k,c}^{u,r}[n] + \xi_{k,m,c}^r[n] \right) \left(\phi_{k,c}^u[n] + \xi_{k,m,c}[n] \right) \\ & \quad \left. - \left(\phi_{k,c}^u[n] - \xi_{k,m,c}[n] \right)^2 \right] \triangleq R_{k,m,c}^{\text{Alb},r}[n]. \end{aligned} \quad (41)$$

Moreover, the right-hand-side (RHS) of constraints (40) is convex w.r.t $\phi_{k',c}^u[n]$. Thus, by applying the first-order Taylor expansion at the given points $\phi_{k',c}^{u,r}[n]$, we can obtain the lower bound $R_{k,m,c}^{\text{Alb},r}[n]$ as in (42), shown at the bottom of the next page.

Therefore, the optimization problem (37) can be approximated by the following problem:

$$\text{(P4.2): } \min_{\mathbf{F}, \Phi, \xi, \eta} \quad \eta \quad (43)$$

$$\begin{aligned} \text{s.t.} \quad & \sum_{m,c,n} \omega_{k,m}[n] \theta_k[n] \left(\frac{\tau_{k,c}^u[n] r_k}{f_k^u[n]} + \frac{\tau_{k,c}^u[n]}{R_{k,m,c}^{\text{Alb},r}[n]} \right) \\ & \leq \eta, \forall k, \end{aligned} \quad (43a)$$

$$\begin{aligned} & \omega_{k,m}[n] \theta_k[n] \sum_c \left(\frac{\tau_{k,c}^u[n] r_k}{f_k^u[n]} + \frac{\tau_{k,c}^u[n]}{R_{k,m,c}^{\text{Alb},r}[n]} \right) \\ & \leq \delta, \quad \forall k, m, n, \end{aligned} \quad (43b)$$

$$E_k^{\text{lo}} + E_k^{\text{os}} + \sum_{m,c,n} \frac{\omega_{k,m}[n] \tau_{k,c}^u[n] P_k^u}{R_{k,m,c}^{\text{Alb},r}[n]} \leq E_k^{\text{max}}, \quad \forall k, \quad (43c)$$

$$\begin{aligned} \xi_{k,m,c}[n] & \leq R_{k,m,c}^{\text{Alb},r}[n], \quad \forall k, m, c, n, \\ & \text{(8), (10), (31g), (31k), (38a),} \end{aligned} \quad (43d)$$

where $\xi = \{\xi_{k,m,c}[n], \forall k, m, c, n\}$.

It can be seen that all constraints are now linear. Hence, problem (43) is a convex optimization problem which can be solved efficiently by any convex optimization solvers such as CVX-Mosek [42]. Specifically, if the sub-channel assignment variables $\phi_{k,c}^u[n], \phi_{k,c}^b$ obtained are all binary, then the relaxation is tight and the obtained solution is also a feasible solution to problem (37). Otherwise, the sub-channel assignment solution needs to be recovered by rounding it to the nearest integer of 0 or 1. Furthermore, since constraints (8) and (10) are met with equalities in the solution of (43), a binary solution can be recovered. The computation complexity of this optimization approach is $\mathcal{O}(I_3 M N C K^{\frac{1}{2}})$, where I_3 is the number of iterations required for the proposed algorithm to converge.

2) *Computation Resource and Sub-Channel Assignment Algorithm (CSA)*: Here, we propose a simplified yet computationally efficient algorithm for computation resource and sub-channel assignments. Recall that our design objective is to minimize the maximum processing time among all GUs and satisfy the energy constraints of individual GUs. Hence, the algorithm enters an iterative sub-channel assignment loop where in each iteration, it searches the GU with the maximum processing time and finds the best sub-channel assignment achieving the smallest and better processing for the underlying GU while reducing the maximum processing time of the system. The algorithm terminates when the maximum processing time of all GUs cannot be reduced further.

Details of the proposed algorithm called “*Computation Resource and Sub-Channel Assignment Algorithm (CSA)*” are given in Algorithm 4. First, we calculate the computation resource for corresponding GUs assigned by each UAV in each time slot as $f_k^u[n] = F_{\text{max}}^u / \sum_k \omega_{k,m}[n], \forall n$. We assign for each GU-BS link, $\phi_{k,c}^b = 1, \forall c \in \mathcal{C}^B$, and also satisfy the constraint (10). Then, in each sub-channel assignment iteration, we find the GU k with the maximum processing time in step 6. This step has the computational complexity of $\mathcal{O}(K)$. For the identified GU k , we can find the best sub-channel assignment in a certain time slot for this user and ensure the maximum processing time of the system not increasing in steps 7–22. After each sub-channel assignment, the maximum processing time of the system is updated and the algorithm terminates when this value cannot be reduced further. The complexity of Algorithm 4 is $\mathcal{O}(L_3 (K + K M N C))$, where L_3 indicates the number of iterations needed to converge.

E. UAV Trajectory Control Sub-Problem

1) *Optimization Approach*: Given $\{\Omega, \Theta, \mathbf{A}, \Upsilon, \mathbf{F}, \Phi\}$, the sub-problem optimizing UAV trajectory control variables $\mathbf{U}$

Algorithm 4 Computation Resource and Sub-Channel Assignment Algorithm (CSA)

Require: M UAVs, K GUs, C sub-channels, T flight time;
 1: **Given:** $\Omega, \Upsilon, \mathbf{A}, \mathbf{F}, \mathbf{U}$;
 2: **Initialization:** Assigning each user link one sub-channel;
Ensure: Min processing time η ;
 3: Calculate $f_k^u[n] = F_m^{\max} / \sum_k \omega_{k,m}[n], \forall n$;
 4: **repeat**
 5: Assign $\phi_{k,c}^b = 1, \forall c \in \mathcal{C}^B$, and satisfy constraint (10);
 6: Find GU $k = \arg\max_{k \in \mathcal{K}} \{T_k\}$;
 7: $\max_{T_k} = T_k$; $\max_{\text{time}} = \max_{k \in \mathcal{K}} \{T_k\}$;
 8: **for** $c = 1 : C^A$ **do**
 9: **for** $m = 1 : M$ **do**
 10: **for** $n = 1 : N$ **do**
 11: **if** $(\omega_{k,m}[n] = 1)$ and $(\omega_{k',m}[n]\phi_{k',c}^u[n] \neq 1)$ **then**
 12: $\phi_{k,c}^u[n] = 1$;
 13: **if** $(T_k \leq \max_{T_k})$ and $(\max_{k \in \mathcal{K}} \{T_k\} \leq \max_{\text{time}})$ and $(E_k \leq E_k^{\max})$ **then**
 14: $\phi_{k,c}^u[n] = 1$;
 15: $\max_{\text{time}} = \max_{k \in \mathcal{K}} \{T_k\}$;
 16: **else**
 17: $\phi_{k,c}^u[n] = 0$;
 18: **end if**
 19: **end if**
 20: **end for**
 21: **end for**
 22: **end for**
 23: $\eta = \max_{k \in \mathcal{K}} \{T_k\}$;
 24: **until** Convergence
 25: Return $\eta, \mathbf{F}, \Phi$.

can be formulated as:

$$\begin{aligned} \text{(P5): } \min_{\mathbf{U}, \eta} \quad & \eta \\ \text{s.t.} \quad & (31c) - (31d), (2) - (5). \end{aligned} \quad (44)$$

This problem is non-convex. We solve it by using the SCA method. To approximate this problem, we introduce auxiliary variables $\gamma_{k,m,c}[n] = R_{k,m,c}^u[n]$, $S_m^{\text{all}}[n] \geq \sum_k S_{k,m}^0[n] + S_m^j[n]$, $S_{k,m}^{0j}[n] \leq \sum_{k' \neq k} \phi_{k',c}[n] S_{k',m}^0[n] + S_m^j[n]$, $\Pi_{k,m}^0[n] \leq H^2 + \|\mathbf{u}_m[n] - \mathbf{g}_k^0\|^2$, and $\Pi_m^j[n] \leq (H - H^j)^2 + \|\mathbf{u}_m[n] - \mathbf{g}^j[n]\|^2$. The approximation processes are similar to that in [26] and [33].

Therefore, this problem can be approximated by:

(P5.1):

$$\min_{\mathbf{U}, \mathbf{F}, \mathbf{S}, \Pi, \eta} \quad \eta \quad (45)$$

$$\text{s.t.} \quad \sum_{m,c,n} \omega_{k,m}[n] \left(\frac{\tau_{k,c}^u[n] r_k}{f_k^u[n]} + \frac{\tau_{k,c}^u[n]}{\gamma_{k,m,c}[n]} \right) \leq \eta, \quad \forall k, \quad (45a)$$

$$\omega_{k,m}[n] \sum_c \left(\frac{\tau_{k,c}^u[n] r_k}{f_k^u[n]} + \frac{\tau_{k,c}^u[n]}{\gamma_{k,m,c}[n]} \right) \leq \delta, \quad \forall k, m, n, \quad (45b)$$

$$E_k^{\text{lo}} + E_k^{\text{os}} + \sum_{m,c,n} \frac{\tau_{k,c}^u[n] P_k^u}{\gamma_{k,m,c}[n]} \leq E_k^{\max}, \quad \forall k, \quad (45c)$$

$$\begin{aligned} \Pi_m^j[n] &\leq (H - H^j)^2 + \|\mathbf{u}_m^r[n] - \mathbf{g}^j[n]\|^2 \\ &+ 2(\mathbf{u}_m^r[n] - \mathbf{g}^j[n])^T (\mathbf{u}_m[n] - \mathbf{u}_m^r[n]), \quad \forall m, n, \end{aligned} \quad (45d)$$

$$\begin{aligned} \Pi_{k,m}^0[n] &\leq H^2 + \|\mathbf{u}_m^r[n] - \mathbf{g}_k^0\|^2 \\ &+ 2(\mathbf{u}_m^r[n] - \mathbf{g}_k^0)^T (\mathbf{u}_m[n] - \mathbf{u}_m^r[n]), \quad \forall k, m, n, \end{aligned} \quad (45e)$$

$$\begin{aligned} d_{\min}^2 &\leq -\|\mathbf{u}_m^r[n] - \mathbf{u}_z^r[n]\|^2 \\ &+ 2(\mathbf{u}_m^r[n] - \mathbf{u}_z^r[n])^T (\mathbf{u}_m[n] - \mathbf{u}_z[n]), \quad \forall z \neq m, n, \end{aligned} \quad (45f)$$

$$\begin{aligned} d_{\min}^2 &\leq \|\mathbf{u}_m^r[n] - \mathbf{g}^j[n]\|^2 \\ &+ 2(\mathbf{u}_m^r[n] - \mathbf{g}^j[n])^T (\mathbf{u}_m[n] - \mathbf{u}_m^r[n]), \quad \forall m, n, \end{aligned} \quad (45g)$$

$$S_m^{\text{all}}[n] \geq \sum_k S_{k,m}^0[n] + S_m^j[n], \quad \forall m, n, \quad (45h)$$

$$S_{k,m}^{0j}[n] \leq \sum_{k' \neq k} \phi_{k',c}[n] S_{k',m}^0[n] + S_m^j[n], \quad \forall m, n, \quad (45i)$$

$$\begin{aligned} \gamma_{k,m,c}[n] &\leq \omega_{k,m}[n] \phi_{k,c}[n] W \left(\log_2 (S_m^{\text{all}}[n] + \sigma^2) \right. \\ &\quad \left. - \log_2 (S_{k,m}^{0j,r}[n] + \sigma^2) - \frac{\log_2(e)}{S_{k,m}^{0j,r}[n] + \sigma^2} \right) \end{aligned}$$

$$\begin{aligned} \log_2 \left(1 + \frac{P_k^u g_{k,m}[n]}{\sum_{k' \neq k} \phi_{k',c}^u[n] P_{k'}^u g_{k',m}[n] + I_m^j[n] + \sigma^2} \right) &\geq \log_2 \left(1 + \frac{P_k^u g_{k,m}[n]}{\sum_{k' \neq k} \phi_{k',c}^{u,r}[n] P_{k'}^u g_{k',m}[n] + I_m^j[n] + \sigma^2} \right) \\ &- \sum_{k' \neq k} A_{k',k,m,c}[n] (\phi_{k',c}^u[n] - \phi_{k',c}^{u,r}[n]) \triangleq R_{k,m,c}^{\text{Au},r}[n], \end{aligned} \quad (42)$$

where

$$A_{k',k,m,c}[n] = \frac{\phi_{k',j}^{u,r}[n] P_k^u P_{k'}^u g_{k',m}[n] g_{k,m}[n] \log_2(e)}{\left(\sum_{k' \neq k} \phi_{k',c}^{u,r}[n] P_{k'}^u g_{k',m}[n] + I_m^j[n] + \sigma^2 \right) \left(\sum_{k' \neq k} \phi_{k',c}^{u,r}[n] P_{k'}^u g_{k',m}[n] + I_m^j[n] + \sigma^2 + P_k^u g_{k,m}[n] \right)}.$$

$$(S_{k,m}^{0j}[n] - S_{k,m}^{0j,r}[n]), \quad (45j)$$

$$P_k^u \rho_0 \geq \Omega_{k,m}^{0,r}[n], P^j \rho_0 \geq \Omega_m^{j,r}[n], \quad (45k)$$

constraints (2), (3),

where $\mathbf{\Gamma} = \{\gamma_{k,m,c}[n], \forall k, m, n, c\}$, $\mathbf{S} = \{S_m^j[n], S_{k,m}^0[n], S_m^{\text{all}}[n], S_{k,m}^{0j}[n], \forall k, m, n\}$, $\mathbf{\Pi} = \{\Pi_m^j[n], \Pi_{k,m}^0[n], \forall k, m, n\}$, $\Omega_{k,m}^{0,r}[n] = \frac{1}{4}[(S_{k,m}^0[n] + \Pi_{k,m}^0[n])^2 + (S_{k,m}^{0,r}[n] - \Pi_{k,m}^{0,r}[n])^2 - 2(S_{k,m}^{0,r}[n] - \Pi_{k,m}^{0,r}[n])(S_{k,m}^0[n] - \Pi_{k,m}^0[n])]$, and $\Omega_m^{j,r}[n] = \frac{1}{4}[(S_m^j[n] + \Pi_m^j[n])^2 + (S_m^{j,r}[n] - \Pi_m^{j,r}[n])^2 - 2(S_m^{j,r}[n] - \Pi_m^{j,r}[n])(S_m^j[n] - \Pi_m^j[n])]$.

Since $\frac{1}{\gamma_{k,m,c}[n]}$ is a convex function w.r.t $\gamma_{k,m,c}[n]$, the objective function is convex. In addition, all constraints are linear. Therefore, problem (45) is a convex problem, which can be solved effectively by using the CVX-Mosek solver [42]. The computation complexity of this optimization approach is $\mathcal{O}(I_4 MNK^{\frac{7}{2}})$, where I_4 is the number of iterations required till convergence.

2) *Circular UAV Trajectory*: The UAVs try to cover and travel close to all GUs to achieve a higher achievable rate, which reduces the energy and offloading time from GUs. Hence, we assume that each UAV serves a circular network partition (i.e., each UAV serves a cluster of GUs located inside a circle with radius r_c (m)). For each network partition served by UAV m , we need to determine its center $\mathbf{c}_m^{\text{init}} = (x_m^{\text{init}}, y_m^{\text{init}})$ and radius d_m^{init} . Given the locations of K GUs, we employ the k-means clustering algorithm [44] to determine the centers of the corresponding network partitions. The radius of the initial circular network partition served by UAV m is given by $d_m^{\text{init}} = \min(d_m^{\text{max1}}, r_c)$. It is assumed that each UAV must remain inside the area of its network partition during the flight period and d_m^{max1} is the maximum radius of the circular UAV trajectory with the same starting and ending point.

To determine d_m^{max1} , we approximate the largest circumference of a circle as the maximum distance, denoted as $D = V_{\text{max}}T$, that the UAV can travel during the flight period. Therefore, we have $d_m^{\text{max1}} \approx D/2\pi$. Let $\varphi_n \triangleq 2\pi \frac{n-1}{N-1}$, $\forall n$, we can initialize $\mathbf{U}^0 = \{\mathbf{u}_m^0[n], \forall m, n\}$ as follows:

$$\mathbf{u}_m^0[n] = (x_m^{\text{init}} + d_m^{\text{init}} \cos \varphi_n, y_m^{\text{init}} + d_m^{\text{init}} \sin \varphi_n), \forall m, n. \quad (46)$$

For the multi-UAV system, $d_{\min}$ denotes the minimum inter-UAV distance to ensure collision avoidance, which must be maintained as we initialize the UAVs' trajectories.

F. Integrated Algorithm

The global solution to the original optimization problem is achieved by the integrated algorithm summarized in Algorithm 5, that alternates solving the aforementioned five sub-problems.

Proposition 1: The objective value of Algorithm 5 monotonically decreases over iterations and this algorithm converges to a feasible solution.

Proof: The proof is given in Appendix I. ■

Algorithm 5 Integrated Algorithm

Require: $\mathcal{K}, \mathcal{M}, \mathcal{C}, W, T$, and locations of GUs;

1: **Initialization:** $\mathbf{\Upsilon}^0, \mathbf{A}^0, \mathbf{F}^0, \mathbf{\Phi}^0, \mathbf{U}^0$;

Ensure: Min processing time (η); Let $r = 1$;

2: **repeat**

3: Solve sub-problem (32) to obtain $\mathbf{\Omega}^r$;

4: Solve sub-problem (35) to obtain $\mathbf{\Theta}^r$;

5: Solve sub-problem (36) to obtain $\mathbf{\Upsilon}^r$ and $\mathbf{A}^r$;

6: Solve sub-problem (43) to obtain $\mathbf{\Phi}^r$ and $\mathbf{F}^r$;

7: Solve sub-problem (45) to obtain $\mathbf{U}^r$;

8: Update $r = r + 1$;

9: **until** Convergence

10: Return $\eta^*, \mathbf{\Omega}^*, \mathbf{\Theta}^*, \mathbf{\Upsilon}^*, \mathbf{A}^*, \mathbf{F}^*, \mathbf{\Phi}^*, \mathbf{U}^*$.

G. Complexity Analysis

In this section, we analyze the complexity of Algorithm 5. In steps 3 to 7, the CVX solver is used to solve the approximated convex problems of the GU association and UAV trajectory control. Since CVX [42] invokes the interior-point method to solve the sub-problems, the involved complexity is $\mathcal{O}(t_1^{1/2}(t_1 + t_2)t_2^2)$, where t_1 is the number of inequality constraints, t_2 denotes the number of variables [43]. The details of the computation complexity of each step are described in each sub-problem. Therefore, let us define I as the number of iterations needed so that Algorithm 5 converges. Hence, the overall computation complexity of Algorithm 5 is $\mathcal{O}(I(MNCK^{\frac{7}{2}}))$.

IV. NUMERICAL RESULTS

We consider different scenarios with a far cloud server from a network serving area with temporary events. The parameter setting for our simulations is similar to that in [25], [40], [41], [45], [46], and [47] and the values of key parameters are summarized in Table II. Specifically, we consider a network area of 1000m×1000m. The task size values s_k are set randomly in the range of [1, 10] Mbits, which is set according to the size of short videos (e.g., video files with the resolution of 30 frames per second (fps) [48]). Simulations are implemented in MATLAB 2022a using the CVX solver on a desktop computer with a 2.10-GHz 12th Gen Intel(R) Core(TM) i7-12700 processor, with 32 GB of 1600-MHz DDR3 memory. Moreover, all simulation results are obtained by averaging over 100 simulation runs, each with different random locations of the GUs.

To investigate the effectiveness of the proposed solutions, we consider the following baseline integrated algorithms:

- **O-Sol1**: It integrates IGA, GSA, COA, CSA algorithms and circular UAV trajectory without jammer attack;
- **J-Sol1**: It integrates IGA, GSA, COA, CSA algorithms and circular UAV trajectory with jammer attack;
- **J-Sol2**: It integrates optimized GU-UAV association, GSA, COA, CSA algorithms and circular UAV trajectory with jammer attack;
- **J-Sol3**: It integrates optimized GU-UAV association, optimized user scheduling, COA, CSA algorithms and circular UAV trajectory with jammer attack;

TABLE II
SIMULATION PARAMETERS

Parameter	Description	Value
M	Number of UAVs	[2, 3]
K	Number of GUs	[10 – 20]
C	Number of sub-channels	[20 – 60]
T	Flight period	[10, 15] s
δ	Length of each time slot	0.5 s
H	Altitude of UAV	100 m
H^j	Altitude of UAV-jammer	100 m
H^b	Altitude of BS	20 m
P_k^u	Transmit power of GU with UAV	26 dBm
P_k^b	Transmit power of GU with BS	28 dBm
P^j	Transmit power of jammer	33 dBm
r_k	Computation resource required for 1-bit input data (CPU cycles/bit)	300
$E_k^{\max}$	Maximum available energy of GU k	1 J
f_k	Computation resource of GU (CPU cycles/s)	2×10^8
$F_m^{\max}$	Maximum computation resource of UAV (CPU cycles/s)	3×10^9
σ^2	Noise power	-174 dBm/Hz
κ	Effective switched capacitance	10^{-28}
ϑ	Communication capacity of the link from the BS to cloud server	[1, 50] Mbps
$d_{\min}$	Minimum inter-UAV distance	20 m
$V_{\max}$	Maximum speed of UAV	20 m/s
ϵ	Small difference of step size	10^{-5}
W	Bandwidth of each sub-channel	1 MHz

 TABLE III
COMPLEXITY OF PROPOSED ALGORITHMS

Algorithms	Complexity	Average Sequence Time ($K = 10, M = 2,$ $C = 20, N = 20$)
O-Sol1	$\mathcal{O}(I_{io}(K + KMNC))$	4.00×10^4
J-Sol1	$\mathcal{O}(I_{ij}(K + KMNC))$	4.45×10^4
J-Sol2	$\mathcal{O}(I_{oa}(MNK^{\frac{7}{2}}))$	2.64×10^5
J-Sol3	$\mathcal{O}(I_{oa}(MNK^{\frac{7}{2}}))$	7.58×10^5
J-Sol4	$\mathcal{O}(I_{oc}(NCK^{\frac{7}{2}}))$	8.22×10^6
J-Sol5	$\mathcal{O}(I_{os}(MNCK^{\frac{7}{2}}))$	1.77×10^7
J-Opt	$\mathcal{O}(I(MNCK^{\frac{7}{2}}))$	2.02×10^7
O-Opt	$\mathcal{O}(I_{oo}(MNCK^{\frac{7}{2}}))$	1.89×10^7

- **J-Sol4**: It integrates optimized GU-UAV association, optimized user scheduling, optimized partial offloading, CSA algorithm, and circular UAV trajectory with jammer attack;
- **J-Sol5**: It integrates optimization approaches but circular UAV trajectory with jammer attack;
- **J-Opt**: Optimization approaches with jammer attack;
- **O-Opt**: Optimization approaches without jammer attack;

These algorithms are based on the iterative algorithm, employing five solutions corresponding to the previously defined five sub-problems. The complexity of the iterative algorithm would be the highest one among these five solutions. As discussed above, the optimization approach for the computation resource and sub-channel assignment sub-problem has the highest complexity of $\mathcal{O}(I_3 MNCK^{\frac{7}{2}})$. This result implies that the complexities of the J-Opt and O-Opt algorithms are similar to that of the J-Sol5 algorithm. However, due to the optimization approach of UAV trajectory control in each iteration of the alternating optimization approach, the J-Opt

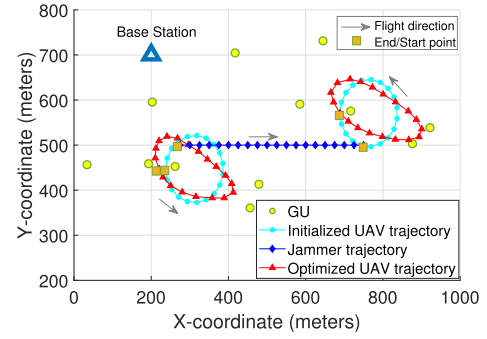
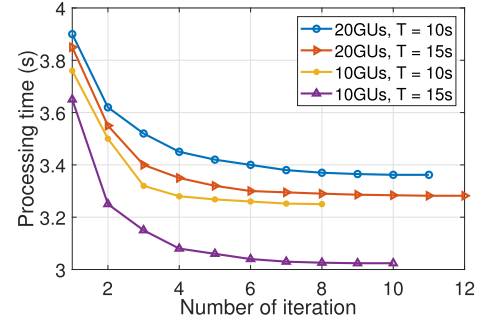


Fig. 2. Optimized UAV trajectories.


 Fig. 3. Convergence of the proposed integrated algorithm (2 UAVs, $H = 100$ m, $C = 20$, $\vartheta = 10$ Mbps).

algorithm requires more iterations to converge. Details of the complexity comparison among these proposed algorithms are presented in Table III, where I_* is the number of iterations needed for the corresponding algorithm to converge. It can be observed that J-Opt has the highest complexity. The performance of proposed algorithms is demonstrated in the following results.

We first study a specific setting with 12 GUs located on the ground and their task size values are set as $s_k = [6, 10, 5, 6, 4, 5, 8, 10, 6, 10, 10, 8]$ Mbits. Fig. 2 illustrates the optimized trajectories of UAVs in the scenarios with 12 GUs, $T = 10$ s, and $C = 20$. This figure shows that UAVs fly close to the GUs and move far away from the jammer.

We now evaluate the effectiveness of the proposed algorithm in comparison with the baselines. The convergence of the integrated Algorithm 5, i.e., J-Opt, is illustrated in Fig. 3. It can be seen that the value of the objective function decreases with the number iterations. Our algorithm converges more slowly to a feasible solution with larger flight period T . This is because the optimization space becomes larger. Moreover, the number of iterations increases as the number of GUs increases due to increasing computation complexity.

Now, we compare the effectiveness of our proposed algorithm with the baselines in Fig. 4. This figure illustrates that J-Opt algorithm achieves the smallest values of the processing time compared to other algorithms. Specifically, the processing time of the optimal design is close to the value of the ideal scenario without jammer attacks, i.e., O-Opt.

Besides, Fig. 5 shows the processing time in the scenario of 2 UAVs and 10 GUs. It can be seen that the processing time for the tasks of GUs decreases as the number of sub-

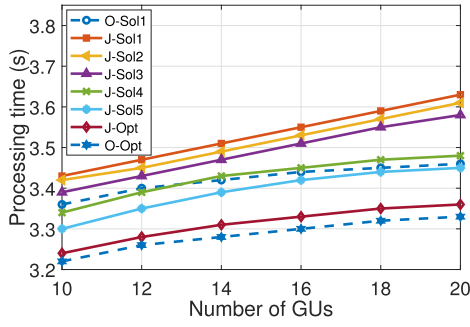


Fig. 4. Processing time for different number of GUs (2 UAVs, $H = 100$ m, $C = 20$, $T = 10$ s, $\vartheta = 10$ Mbps).

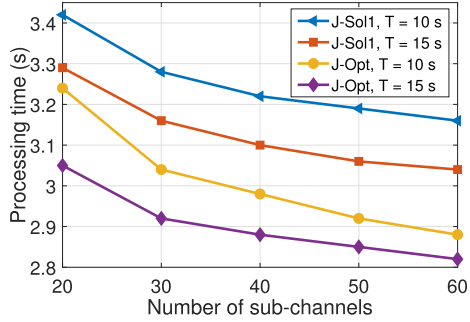


Fig. 5. Processing time for different number of sub-channels (2 UAVs, $H = 100$ m, $K = 10$ GUs, $\vartheta = 10$ Mbps).

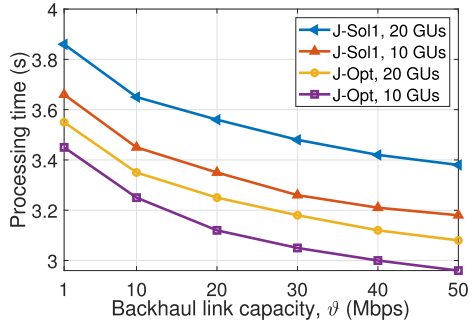


Fig. 6. Processing time for different backhaul link capacity, ϑ (Mbps), (2 UAVs, $H = 100$ m, $C = 20$, $T = 10$ s).

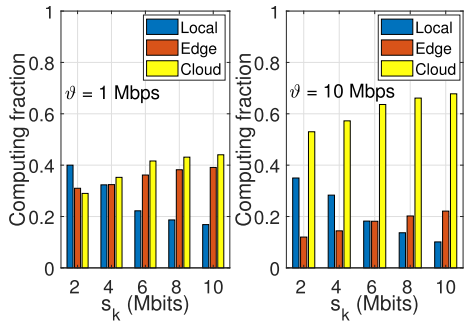


Fig. 7. Computation distribution with different task size values (2 UAVs, $H = 100$ m, $K = 10$ GUs, $C = 20$, $T = 10$ s).

channels increases. This is reasonable because the larger number of sub-channels allows larger transmission rates and hence larger fractions of computation load to be offloaded, achieving processing time savings for both the edge servers and GUs. Moreover, it can also be seen that our proposed

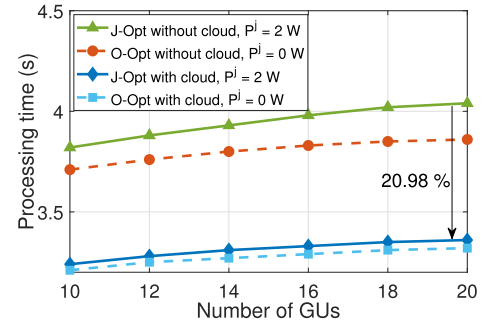


Fig. 8. Processing time for different number of GUs and different P^j values (2 UAVs, $H = 100$ m, $C = 20$, $T = 10$ s, $\vartheta = 10$ Mbps).

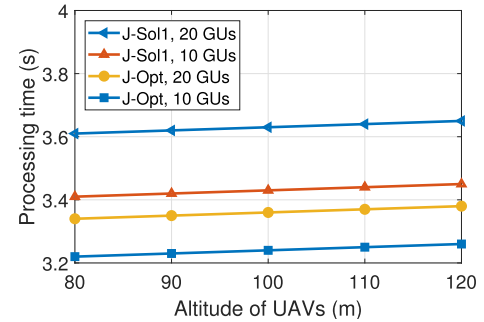


Fig. 9. Processing time for different altitudes of UAVs, H (m), (2 UAVs, $C = 20$, $T = 10$ s, $\vartheta = 10$ Mbps).

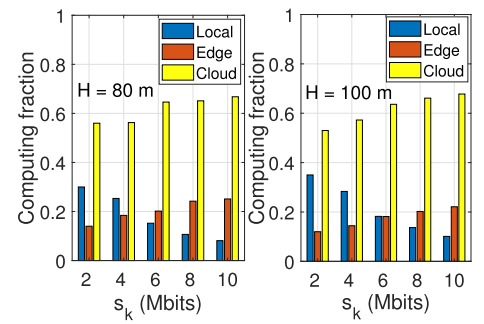


Fig. 10. Computation distribution with different task size values (2 UAVs, $K = 10$ GUs, $C = 20$, $T = 10$ s, $\vartheta = 10$ Mbps).

optimal design can save processing time compared to the baseline algorithms. These results indicate the effectiveness of our proposed algorithm in solving the joint optimization problem.

Fig. 6 illustrates the processing time for different backhaul link capacities in comparison of our proposed algorithm J-Opt with J-Sol1 in case of 2 UAVs, $C = 20$, $T = 10$ s. It can be seen that the processing time of the system decreases as the backhaul link capacity increases. This is because with higher backhaul link capacity more sub-tasks of GUs would be uploaded to process at the cloud server. Details of the computation load distribution are illustrated in Fig. 7 over network layers with our proposed algorithm, i.e., J-Opt.

Fig. 8 describes the processing time with different numbers of GUs and different P^j values. It can be seen that the processing time for the tasks of GUs increases as the number of GUs increases. This is because, given the bandwidth and computation resources, the transmission rates of GUs decrease

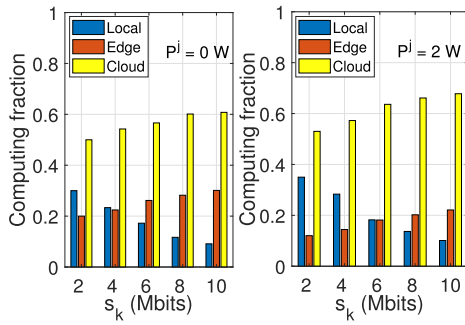


Fig. 11. Computation distribution with different task size values (2 UAVs, $H = 100$ m, $K = 10$ GUs, $C = 20$, $T = 10$ s, $\vartheta = 10$ Mbps).

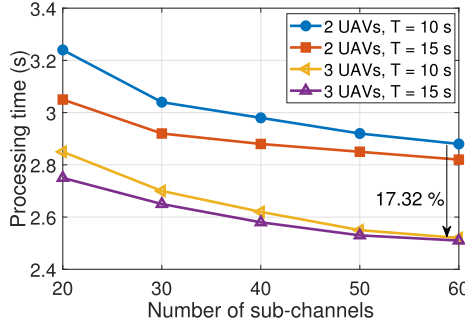


Fig. 12. Processing time for different number of sub-channels ($K = 10$ GUs, $H = 100$ m, $\vartheta = 10$ Mbps).

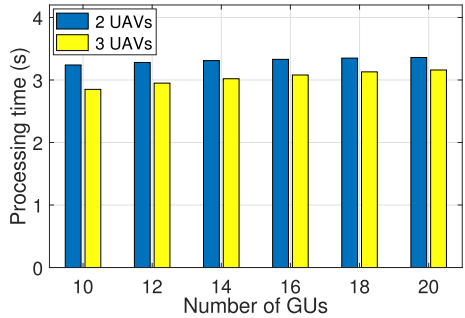


Fig. 13. Processing time for different number of GUs ($T = 10$ s, $H = 100$ m, $C = 20$, $\vartheta = 10$ Mbps).

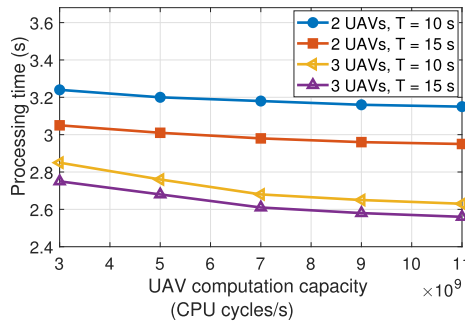


Fig. 14. Processing time for different computation capacities of UAVs (CPU cycles/s) ($C = 20$, $K = 10$ GUs, $H = 100$ m, $\vartheta = 10$ Mbps).

with a larger number of served GUs. Hence, it takes more time to process completely the sub-tasks at the local, edge, or cloud server. Also, we see that the presence of a jammer results in higher processing time, and that the utilization of a distant cloud server reduces the processing time. For instance,

it could reduce 20.98% in the case of 20 GUs. Moreover, the gap between the scenarios with and without jammer attacks is smaller with the utilization of a cloud server.

Fig. 9 shows the performance in a simulation conducted with 2 UAVs, where the UAVs' altitudes are randomly chosen within a range of 80 m to 120 m. It can be observed that the average max-min processing time slightly increases as the altitude of UAVs increases. This can be attributed to the fact that, at higher UAV altitudes, the achievable rate between GUs and UAVs decreases. Consequently, the transmission time required to offload sub-tasks to the edge server increases. As a result, the processing time for sub-tasks at the GU, edge, or cloud servers becomes longer. The details of computation load distribution across layers are illustrated in Fig. 10. It is observed that, at higher UAV altitudes, the portion of sub-tasks offloaded to the edge server decreases, while larger sub-tasks are computed locally.

Fig. 11 illustrates the computation load distribution over network layers. This figure shows that larger task size values lead to less computation load distributed at the GUs, while the scenarios without jammer attacks result in a higher computation load to be processed at the edge servers.

Furthermore, Fig. 12 shows that the processing time decreases as the number of sub-channels increases and that deploying 3 UAVs can reduce the maximum processing time among all GUs compared to that with 2 UAVs. Specifically, it could reduce 17.32% in the case of 60 sub-channels and $T = 10$ s. Meanwhile, Fig. 13 illustrates that deploying 3 UAVs can reduce the maximum processing time among all GUs compared to those with 2 UAVs.

Finally, Fig. 14 shows that the processing time for the tasks of GUs decreases as higher computation capacity of UAVs. This is due to the fact that a stronger computing capability of UAVs can compute faster the offloaded sub-tasks from the GUs.

V. CONCLUSION

In this paper, we have studied the joint optimization of GU-UAV association, partial offloading control and bit allocation, computation resource, sub-channel assignment, and UAV trajectory control against an active aerial attacker for UAV-based wireless networks. We have employed the alternating optimization approach to solve the underlying problem and leveraged the SCA method to tackle the non-convex sub-problems of computation resource and sub-channel assignment and UAV trajectory control. Moreover, we proposed computationally efficient algorithms to solve with lower complexity the sub-problems of GU-UAV association, partial offloading control and bit allocation, and computation resource and sub-channel assignment. Numerical results have been provided to demonstrate the effectiveness of the proposed algorithms compared to the baselines, showing the strong impacts of various system parameters such as the numbers of GUs and sub-channels as well as the data size values on the computation distribution and processing time for individual GUs' tasks. In our future work, we will investigate highly dynamic systems that incorporate flexible 3D UAV trajectory control, UAV CSI estimation, smart jammers, low-power design, and the

partitioning of offloading sub-tasks. To address these challenging problems, we plan to investigate simpler machine learning-based techniques and compare their performance with optimization-based approaches.

APPENDIX I PROOF OF PROPOSITION 1

Here, we prove that the objective value of Algorithm 5 is non-increasing after each iteration and Algorithm 5 converges to a feasible solution.

Let $\Omega^r, \Theta^r, \Upsilon^r, \mathbf{A}^r, \mathbf{F}^r, \Phi^r$, and $\mathbf{U}^r$ denote the feasible solution at r -th iteration of the approximation process. For step 3 of Algorithm 5, since the optimal solution of (P1) is obtained with given $\Theta^r, \Upsilon^r, \mathbf{A}^r, \mathbf{F}^r, \Phi^r$, and $\mathbf{U}^r$, we have

$$\begin{aligned} \eta(\Omega^r, \Theta^r, \Upsilon^r, \mathbf{A}^r, \mathbf{F}^r, \Phi^r, \mathbf{U}^r) \\ \geq \eta(\Omega^{r+1}, \Theta^r, \Upsilon^r, \mathbf{A}^r, \mathbf{F}^r, \Phi^r, \mathbf{U}^r). \end{aligned} \quad (47)$$

In addition, for given $\Omega^{r+1}, \Upsilon^r, \mathbf{A}^r, \mathbf{F}^r, \Phi^r$, and $\mathbf{U}^r$ obtained in step 3 of Algorithm 5, it follows that

$$\begin{aligned} \eta(\Omega^{r+1}, \Theta^r, \Upsilon^r, \mathbf{A}^r, \mathbf{F}^r, \Phi^r, \mathbf{U}^r) \\ \geq \eta(\Omega^{r+1}, \Theta^{r+1}, \Upsilon^r, \mathbf{A}^r, \mathbf{F}^r, \Phi^r, \mathbf{U}^r). \end{aligned} \quad (48)$$

This is because (P2) is optimally solved to obtain Θ^{r+1} . Moreover, we obtain

$$\begin{aligned} \eta(\Omega^{r+1}, \Theta^{r+1}, \Upsilon^r, \mathbf{A}^r, \mathbf{F}^r, \Phi^r, \mathbf{U}^r) \\ \geq \eta(\Omega^{r+1}, \Theta^{r+1}, \Upsilon^{r+1}, \mathbf{A}^{r+1}, \mathbf{F}^r, \Phi^r, \mathbf{U}^r). \end{aligned} \quad (49)$$

This is because problem (P3) is optimally solved to obtain Υ^{r+1} and $\mathbf{A}^{r+1}$. Next, for given $\Omega^{r+1}, \Theta^{r+1}, \Upsilon^{r+1}, \mathbf{A}^{r+1}, \mathbf{F}^r, \Phi^r$, and $\mathbf{U}^r$ achieved in step 5 of Algorithm 5, we also have

$$\begin{aligned} \eta(\Omega^{r+1}, \Theta^{r+1}, \Upsilon^{r+1}, \mathbf{A}^{r+1}, \mathbf{F}^r, \Phi^r, \mathbf{U}^r) \\ \geq \eta(\Omega^{r+1}, \Theta^{r+1}, \Upsilon^{r+1}, \mathbf{A}^{r+1}, \mathbf{F}^{r+1}, \Phi^{r+1}, \mathbf{U}^r). \end{aligned} \quad (50)$$

This result is obtained by solving problem (P4.2) based on the SCA method. Finally, for given $\Omega^{r+1}, \Theta^{r+1}, \Upsilon^{r+1}, \mathbf{A}^{r+1}, \mathbf{F}^{r+1}, \Phi^{r+1}$, and $\mathbf{U}^r$ obtained in step 6 of Algorithm 5, we have

$$\begin{aligned} \eta(\Omega^{r+1}, \Theta^{r+1}, \Upsilon^{r+1}, \mathbf{A}^{r+1}, \mathbf{F}^{r+1}, \Phi^{r+1}, \mathbf{U}^r) \\ \geq \eta(\Omega^{r+1}, \Theta^{r+1}, \Upsilon^{r+1}, \mathbf{A}^{r+1}, \mathbf{F}^{r+1}, \Phi^{r+1}, \mathbf{U}^{r+1}). \end{aligned} \quad (51)$$

This result holds since the problem (P5.1) achieves the solutions $\mathbf{U}^{r+1}$ by applying the SCA method. Using the results in (47)–(51), we obtain

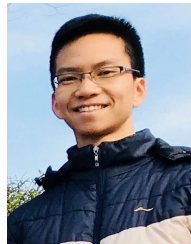
$$\begin{aligned} \eta(\Omega, \Theta, \Upsilon, \mathbf{A}, \mathbf{F}, \Phi, \mathbf{U}) \\ \geq \eta(\Omega^{r+1}, \Theta^{r+1}, \Upsilon^{r+1}, \mathbf{A}^{r+1}, \mathbf{F}^{r+1}, \Phi^{r+1}, \mathbf{U}^{r+1}), \end{aligned} \quad (52)$$

which indicates that the objective value of problem (P) is non-increasing after each iteration of Algorithm 5. Since the objective value of problem (P) is lower bounded by a finite value, Algorithm 5 is guaranteed to converge to a feasible solution. This completes the proof.

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